

Feasibility Analysis & MVP Roadmap

Project: Climate Response Hub

Team: Program Counter

1. Executive Summary

Currently, emergency coordinators at Toronto Emergency Management (TEM) and Toronto Shelter & Support Services (TSSS) face significant cognitive load during extreme heat events, as they must mentally connect fragmented data across multiple tabs, such as the Heat Vulnerability Index and the Cool Spaces Near You map.

Climate Response Hub is a decision-support dashboard that integrates climate risk data and shelter access gaps into a single, unified view. By enabling coordinators to log their dispatch actions and track real-time status, the platform serves as a collaborative tool to prevent duplicate deployments and maximize the efficiency of relief resources.

2. Feasibility Analysis

This solution meets the practical business and technical feasibility criteria required by the hackathon, directly addressing the goals of municipal and utility partners.

A. Operational Feasibility

Preventing Duplicate Efforts: When a coordinator logs an action (e.g., dispatching a mobile cooling unit) on the dashboard, the neighborhood is immediately marked as 'handled'. This transparent tracking eliminates confusion, double deployments, and wasted resources among different departments.

Reduced Decision-Making Time: By eliminating the "tab-switching" cognitive load, the platform accelerates crisis triage, transforming hours of manual data cross-referencing into seconds of actionable insights.

B. Economic Feasibility

Resource Optimization: Efficiently targets limited physical relief resources to high-risk areas with the highest vulnerability scores and severe accessibility gaps, reducing unnecessary emergency service dispatches.

Cost-Effective Architecture: Leverages existing public data ecosystems (e.g., the City of Toronto's Heat Relief Network) and Esri's infrastructure, requiring very low initial capital for deployment.

C. Technical Feasibility

Verified Data Integration: Utilizes the robust Esri ArcGIS Living Atlas for real-time climate vulnerability and demographic layers, ensuring high analytical accuracy.

Flexible Data Strategy (Real + Mock): While climate risks use real data, the platform demonstrates its capability to handle real-time infrastructure status by using Mock APIs for cooling center capacity and backup power visibility. The Node.js/Express and MongoDB architecture is designed to easily swap these mock endpoints with live APIs from utility partners in the future.

3. MVP Scope & 6-Month Roadmap

Instead of overbuilding, our roadmap focuses on a minimum viable product that immediately solves the core collaboration problem, followed by strategic expansions.

Phase 1: Current MVP (Months 0-1)

Integration of Esri GIS Heat Vulnerability Index mapping.

Visualization of 500+ Cool Spaces with mock data simulating live capacity and backup power status.

Implementation of the collaborative CRUD workflow (Log Action Plan & Track Status: Pending → Deployed → Completed).

Phase 2: Data & Partner Integration (Months 2-3)

Transition from mock infrastructure data to actual utility APIs (e.g., Alectra Utilities) to monitor live power grid and backup generator statuses during extreme weather events.

Integration with internal municipal dispatch systems (e.g., Toronto Water) for automated physical fleet coordination.

Phase 3: Service Expansion (Months 4-6)

Expansion beyond heatwaves to address winter cold snaps by mapping emergency Warming Centres.

Development of automated Energy Equity reporting modules to assist utility partners with long-term infrastructure upgrade planning.

4. Strategic Impact

The Climate Response Hub goes beyond a simple monitoring tool; it serves as a critical infrastructure piece to achieve national and corporate resilience targets.

Alignment with Government of Canada Goals: Directly supports the national mandate to implement evidence-based adaptation measures and eliminate extreme heatwave deaths by 2040.

Supporting Alectra Utilities: By providing unprecedented visibility into backup power and community vulnerability, the platform actively supports Alectra's climate resilience and emergency preparedness goals.